**Table 4.4 Elements and their symbols with their atomic number and mass number and valency.**

| Element | Symbol | Atomic Number | Mass Number | Valency |
|---|---|---|---|---|
| Hydrogen | H | 1 | 1 | 1 |
| Carbon | C | 6 | 12 | 4 |
| Oxygen | O | 8 | 16 | 2 |
| Sodium | Na | 11 | 23 | 1 |
| Calcium | Ca | 20 | 40 | 2 |

**What makes atoms stick together?**

Electrons carry a negative electric charge, and protons carry a positive charge. The attraction between them holds electrons in orbits.

Valency is the combining property of an atom. It is a measure of how many hydrogen atoms it can combine with. For example, oxygen can combine with two hydrogen atoms and create water molecule. So, the valency of oxygen atom is two. In the case of chlorine, it can combine with only one hydrogen to create HCl (hydrochloric acid). Here, the valency of chlorine is one. Methane ($CH_4$) has one carbon atom combining with four hydrogen atoms. Can you guess the valency of carbon in methane? In ammonia molecule, nitrogen combines with three hydrogen atoms. What is the valency of nitrogen in ammonia?

Atoms of different elements combine with each other to form molecules. Valency determines the number of atoms of an element that combines with atom or atoms of another type.

The element having valency one is called monovalent. Example: Hydrogen and Sodium. The elements having valency two are called divalent. Example: Oxygen and Beryllium. The elements having valency three are called trivalent. Example: Nitrogen and Aluminium. Some elements exhibit more than one valency. For example, iron combines with oxygen to form two types of oxides namely, ferrous oxide (exhibits valency 2) and ferric oxide (exhibits valency 3). We will study about them in detail later.

When atoms of different elements combine with each other, molecules of compounds are formed. In these instances, it is necessary to know the valancies of those elements. Valencies of some elements are given in Table 4.4.

## Points to Remember

- An atom is the smallest particle of an element that retains its chemical properties. They are very tiny compared to other particles.
- Atoms are too small to be seen by the naked eye or even through microscope.
- An atom consists mostly of empty space.
- Atoms of same element are identical, and atoms of different elements differ.
- An atom consists of a dense nucleus which has positively-charged protons and electrically-neutral neutrons.
- The protons and neutrons are called nucleons.
- An atom is electrically neutral. They contain equal number of protons and electrons.
- Atomic number is the number of protons in an atom.
- The total number of protons and neutrons present in the nucleus of an atom is called its mass number.
- Valency is defined as the combining capacity of an element.



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